Huawei Access Node QoS Congestion Event Standard Operating Procedure (ALARM_1020)

Node: JAY-ACC-77 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1020

Description

The Huawei Access Node QoS Congestion Event (ALARM_1020) is triggered when the QoS configuration is unable to manage the excessive traffic on the access node, causing microbursts and interface tail drops, resulting in QoS queue starvation. This phenomenon is rooted in the underlying protocol mechanics of the Huawei access node, where the QoS configuration is not effectively prioritizing traffic, leading to congestion. The root cause of this issue is often due to misconfiguration of the QoS policy, inadequate bandwidth allocation, or hardware limitations. As a Senior MTS, it is essential to promptly diagnose and resolve this issue to prevent further congestion and maintain optimal network performance.

Pre-requisite Checks

- Verify that the alarm is triggered on the Huawei access node (JAY-ACC-77) and not on any downstream or upstream devices.

- Check the redundant link status to ensure that the link is active and not in a failed state.

- Verify that the QoS configuration is correctly applied and not overridden by any other configuration.

- Gather the show commands output to analyze the QoS configuration and traffic statistics.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy	High
Inadequate bandwidth allocation	Medium
Hardware limitations (buffer size, queue depth)	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration system qos, Expected Output: Current QoS configuration, including policy maps, classes, and queues., Description: This command displays the current QoS configuration, allowing us to verify that the configuration is correctly applied and not overridden by any other configuration.

Command: display interface brief, Expected Output: Interface statistics, including errors, discards, and drop counts., Description: This command displays the interface statistics, allowing us to identify any signs of congestion or errors on the access node.
```

```
Command: display qos statistics, Expected Output: QoS statistics, including drop counts, queue lengths, and policy hit counts., Description: This command displays the QoS statistics, allowing us to verify that the QoS configuration is effectively managing the traffic and preventing congestion.
```

Resolution Steps

Step 1: Step: Verify QoS configuration, Command: display current-configuration system qos, Description: Verify that the QoS configuration is correctly applied and not overridden by any other configuration.

Step 2: Step: Adjust QoS policy, Command: system-qos policy-map policy-name class class-name queue queue-name, Description: Adjust the QoS policy by modifying the policy maps, classes, and queues to ensure that traffic is correctly prioritized and managed.

Step 3: Step: Check bandwidth allocation, Command: display interface brief, Description: Verify that the bandwidth allocation is adequate and not causing congestion on the access node.

Post-Check Verification

```
Command: display qos statistics, Expected Output: QoS statistics, including drop counts, queue lengths, and policy hit counts., Description: Verify that the QoS configuration is effectively managing the traffic and preventing congestion.

Command: display interface brief, Expected Output: Interface statistics, including errors, discards, and drop counts., Description: Verify that the interface statistics are within normal ranges and there are no signs of congestion or errors on the access node.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: system-qos policy-map policy-name class class-name queue queue-name, Description: Restore the original QoS configuration to prevent further congestion and maintain optimal network performance.

- Step: Verify QoS configuration, Command: display current-configuration system qos, Description: Verify that the QoS configuration is correctly applied and not overridden by any other configuration.